

(4.)

$$T_0(x_0, y_0)$$

 $\vdots$

$$T_n(x_n, y_n)$$

$$x_i \neq x_j \text{ i.e. } i \neq j$$

pokaži, da obstaja nat. 1 polinom,
ki poteka skozi vse točke

$$p(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

$$a_0 + a_1 x_0 + \dots + a_n x_0^n = y_0$$

 $\vdots$

$$a_0 + a_1 x_n + \dots + a_n x_n^n = y_n$$

$$\underbrace{\begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix}}_{\text{Vandermondeova matrica}} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Vandermondeova matrica = V

če $\det V \neq 0 \Leftrightarrow$ enolična rešitev

$$\begin{bmatrix} 1 & x_0 & \dots & x_0^{n-1} & x_0^n \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & x_n & \dots & x_n^{n-1} & x_n^n \end{bmatrix} \sim \begin{bmatrix} 1 & x_0 & \dots & 0 & x_0^{n-1}(x_1 - x_0) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & x_n & \dots & x_n^{n-1}(x_n - x_0) \end{bmatrix} \sim \dots \sim \begin{bmatrix} 1 & 0 & \dots & 0 & 0 \\ 1 & (x_1 - x_0) & \dots & x_1^{n-2}(x_1 - x_0) & x_1^{n-1}(x_1 - x_0) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & (x_n - x_0) & \dots & x_n^{n-2}(x_n - x_0) & x_n^{n-1}(x_n - x_0) \end{bmatrix}$$

$$\det V_n = 1 \cdot \det \begin{bmatrix} x_1 - x_0 & x_1(x_1 - x_0) & \dots & x_1^{n-1}(x_1 - x_0) \\ \vdots & \vdots & & \vdots \\ x_n - x_0 & \dots & \dots & x_n^{n-1}(x_n - x_0) \end{bmatrix} = (x_1 - x_0)(x_2 - x_0) \dots (x_n - x_0) \det V_{n-1} =$$

$$= \underbrace{(x_1 - x_0)(x_2 - x_0) \dots (x_n - x_0)}_{\text{...}} \underbrace{(x_2 - x_1)(x_3 - x_1) \dots (x_n - x_1)}_{\text{...}} \underbrace{(x_3 - x_2)(x_4 - x_2) \dots (x_n - x_2)}_{\text{...}} \dots$$

$$\boxed{= \prod_{i > j} (x_i - x_j) \neq 0 \Rightarrow \text{rešitev je enoličen (rank } V = \text{rank} \left(\begin{bmatrix} y_0 \\ \vdots \\ y_n \end{bmatrix} \right) \Rightarrow \text{sistem ima eno rešitev)}}$$

5. Z uporabo Cramerjevega izračunaj inverz

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 2 & -4 \\ 2 & -1 & 0 \end{bmatrix}$$

$$\det A = -16 + 9 + 12 - 4 = 1$$

$$\text{co}(A) = \begin{bmatrix} -4 & -8 & -7 \\ 3 & 6 & 5 \\ -2 & -5 & -4 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} -4 & 3 & -2 \\ -8 & 6 & -5 \\ -7 & 5 & -4 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

6. s Cramerjem reši sist. enačb:

$$2x_1 + x_2 + x_3 = 1$$

$$3x_1 + x_3 = 4$$

$$x_1 - x_2 - x_3 = 2$$

$$\det \begin{bmatrix} 2 & 1 & 1 \\ 3 & 0 & 1 \\ 1 & -1 & -1 \end{bmatrix} = 1 - 3 + 2 + 3 = 3$$

$$x_1 = 1$$

$$x_2 = -2$$

$$x_3 = 1$$

$$\det A \neq 0, B = \begin{bmatrix} b_1 \\ \vdots \end{bmatrix}$$

$$x_i = \frac{\det A^{(i)}}{\det A}, A^{(i)} \rightarrow i\text{-ti stolpec}$$

enačbe zamenjamo z desno stranjo enačbe

$$A^{(1)} = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 0 & 1 \\ 2 & -1 & -1 \end{bmatrix}$$

$$\det A^{(1)} = 2 - 4 + 1 + 4 = 3$$

$$A^{(2)} = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 4 & 1 \\ 1 & 2 & -1 \end{bmatrix}$$

$$\det A^{(2)} = -8 + 1 + 6 - 4 - 4 + 3 = -6$$

$$A^{(3)} = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 0 & 4 \\ 1 & -1 & 2 \end{bmatrix} \rightarrow \det A^{(3)} = 4 - 3 + 8 - 6 = 3$$

⑦ poddet. $\rightarrow$ izračunaj $\text{rk}(A)$

$$A = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}; x, y, z \in \mathbb{R}$$

$\text{rk} A =$ velikost najv. nenulne poddet.

$$\det A = -xyz + xyz = 0$$

$$2 \times 2: \begin{matrix} z^2, & z \cdot x, & -yz, & z \cdot x, & x^2, & -xy, & -yz, & y^2, & xy \\ & \downarrow & & & & & & & \\ x^2, & y^2, & z^2, & -xy, & xz, & -yz \end{matrix}$$

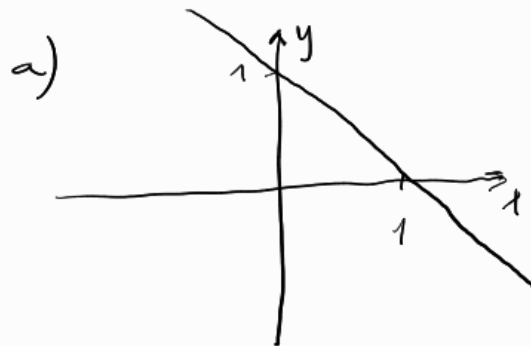
$$\text{rk} A = 2 \Leftrightarrow x \neq 0 \vee y \neq 0 \vee z \neq 0$$

$$\text{rk} A = 0 \Leftrightarrow x = y = z = 0$$

Vekt. prostor

① Katere množice so vekt. podprostorji euklidskega prostora?

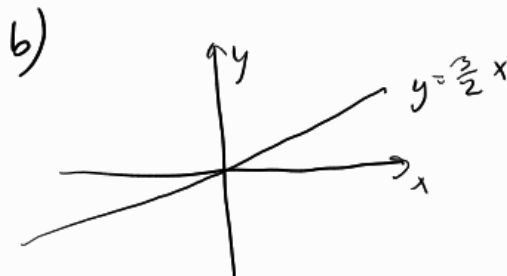
a) $\{(x, y) \in \mathbb{R}^2 \mid x + y = 1\}$



$$y = -x + 1$$

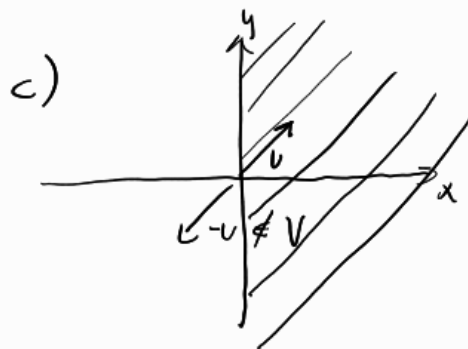
če $x = 0$: $\forall v \in V$: $\alpha v = 0 \notin V$
Ne.

b) $\{(x, y) \in \mathbb{R}^2 \mid 3x - 2y = 0\}$

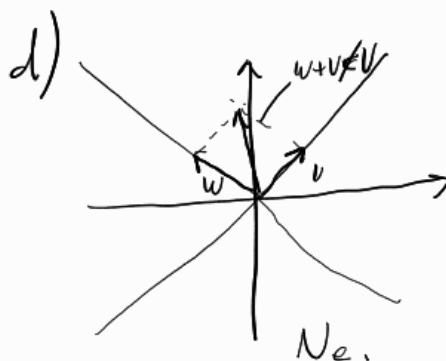


Da.

c) $\{(x, y) \in \mathbb{R}^2 \mid x \geq 0\}$



Ne.



$$x = \pm y$$

Ne.